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Moving Mechanical Assemblies for Space and Launch Vehicles

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1 Scope

1.1 Purpose

This specification sets forth the general requirements for the design, manufacture, quality control, and testing of moving mechanical assemblies to be used on space and launch vehicles.

1.2 Application

This specification is intended for reference in vehicle or equipment specifications, to incorporate those requirements which are common to most moving mechanical assemblies for space and launch vehicles. The requirements stated are a composite of those that have been found to be cost effective for high reliability space and launch vehicle applications. The requirements covered by this specification are applicable to the mechanical or electromechanical devices that control the movement of a mechanical part of a space or launch vehicle relative to another part. They include, but are not limited to, deployment mechanisms, sensor mechanisms, pointing mechanisms, drive mechanisms, despin mechanisms, separation mechanisms, momentum and reaction wheels, control moment gyros, gimbals, and other mechanisms required to perform specific functions. The requirements apply to the overall moving mechanical assembly as well as to the actuators, motors, springs, dampers, linkages, latches, bearings, clutches, cams, booms, gimbals, slip rings, gears, and instrumentation that are an integral part of these mechanical assemblies. When this specification is used as a reference document to specify general requirements for moving mechanical assemblies to be used on launch vehicles, injection stages, reentry vehicles, or other vehicles, the term "space vehicle" is to be interpreted as the applicable vehicle.

2 Applicable Documents

2.1 Government Documents

2.1.1 Specifications, Standards, and Handbooks

The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS:

Federal

QQ-N-290 Nickel Plating (Electrodeposited)

QQ-C-320 Chromium Plating (Electrodeposited)

QQ-S-763 Steel Bars, Shapes, and Forgings, Corrosion Resisting

Military

MIL-M-3171 Magnesium Alloy, Processes for Pretreatment - and Prevention of Corrosion on

MIL-C-5541 Chemical Conversion Coatings on Aluminum and Aluminum Alloys

MIL-H-5606 Hydraulic Fluid, Petroleum Base; Aircraft Missile and Ordnance
 MIL-H-6083 Hydraulic Fluid; Petroleum Base: For Preservation & Operation
 MIL-H-6088 Heat Treatment of Aluminum Alloys
 MIL-I-6868 Inspection Process, Magnetic Particle
 MIL-H-6875 Heat Treatment of Steel (Aircraft Practice), Process for
 MIL-F-7179 Finishes and Coatings, General Specification for Protection of Aerospace
 Weapons Systems, Structures and Parts
 MIL-S-7742 Screw Threads, Standard, Optimum Selected Series, General Specification
 for
 MIL-M-8609 Motors, Direct Current, 28V System, Aircraft, General Specification For
 MIL-A-8625 Anodic Coatings, for Aluminum and Aluminum Alloys
 MIL-S-8879 Screw Threads, Controlled Radius Root with Increased Minor Diameter,
 General Specification for
 DOD-E-8983 Electronic Equipment, Aerospace, Extended Space Environment, General
 Specification for
 MIL-S-13572 Spring, Helical, Compression and Extension
 MIL-M-13786 Motors, Fractional Horsepower, Direct Current and Universal (For Com-
 munication and Other Electronic and Special Military Applications)
 MIL-M-13787 Motors, Alternating Current, Fractional Horsepower, Squirrel Cage (For
 Communication and Other Electronic and Special Military Applications)
 MIL-C-26074 Coatings, Electroless Nickel, Requirements
 MIL-M-45202 Magnesium Alloys, Anodic Treatment of
 MIL-H-81200 Heat Treatment of Titanium and Titanium Alloys
 MIL-B-81793 Bearing, Ball, Precision, for Instruments and Rotating Components
 DOD-W-83575 Wiring Harness, Space Vehicle, Design and Testing, General Specification
 for
 DOD-E-83578 Explosive Ordnance for Space Vehicles, General Specification for
 STANDARDS:
 Federal:
 FED-STD-209 Clean Room and Work Station Requirements, Controlled Environment
 Military:
 MIL-STD-29 Springs, Mechanical: Drawing Requirements for
 MIL-STD-453 Radiographic Inspection Methods
 MIL-STD-889 Dissimilar Metals
 MIL-STD-1246 Product Cleanliness Levels and Contamination Control Program
 MIL-STD-1515 Fasteners for Use in the Design and Construction of Aerospace Mechanical
 Systems
 MIL-STD-1522 Standard General Requirements for Safe Design and Operation of Pres-
 surized Missile and Space Systems
 MIL-STD-1540 Test Requirements for Space Vehicles
 MIL-STD-1547 Electronic Parts, Materials and Processes for Space and Launch Vehicles
 MIL-STD-1568 Materials and Processes for Corrosion Prevention and Control in Aerospace
 Systems
 MIL-STD-6866 Inspection, Liquid Penetrant
 MS-33540 Safety Wiring and Cotter Pinning, General Practices for

HANDBOOKS:

MIL-HDBK-5 Metallic Materials and Elements for Aerospace Vehicle Structures

MIL-HDBK-17 Plastics for Aerospace Vehicles - Part 1, Reinforced Plastics; - Part II, Transparent Glazing Materials

MIL-HDBK-23 Structural Sandwich Composites

MIL-HDBK-217 Reliability Prediction of Electronic Equipment

DOD-HDBK-263 Electrostatic Discharge Control Handbook for Protection of Electrical b Electronic Parts, Assemblies and Equipment

2.1.2 Other Government Documents, Drawings, and Publications

The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

SP-R-0022 Vacuum Stability Requirements of Polymeric Materials for Spacecraft Applications (NASA JSC)

MSFC-SPEC-522 Stress Corrosion Cracking Control (NASA MS F C)

NHB 1700.7 Safety Policy and Requirements for Payloads Using the Space Transportation System (STS) (NASA)

(Copies of specifications, standards, handbooks, drawings, publications, and other government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Other Publications

The following document(s), form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA Handbook American Gear Manufacturer's Association Handbook

(Application for copies should be addressed to AGMA, 1500 King St., Suite 201, Alexandria, VA 22314)

ANTI-FRICTION BEARING MANUFACTURERS ASSOCIATION (AFBMA)

AFBMA Standards AFBMA Standards for Ball and Roller Bearings and Balls

(Application for copies should be addressed to AFBMA, Suite 700, 1101 Connecticut Ave.N.W., Washington, D.C. 20036)

(Industry association specifications and standards are normally available for reference from libraries. They are also distributed among technical groups and using Federal agencies)

3 Requirements

3.1 Requirement Weighting Factors

The requirements stated herein are a composite of the designs, items, and practices found to be cost effective for high reliability moving mechanical assemblies used in space vehicles. Because of the broad scope of the requirements stated herein and the wide range of applications, all requirements stated are not of equal importance or weight. They have been divided into four categories of importance, ranging from requirements that are imposed on all applications to examples of acceptable designs, items, and practices. The relative weighting of requirements is an important consideration when tailoring the specification to specific applications and in making trade studies of alternatives (see 6.1). The weighting factors that are incorporated in the specification are:

a. Weighting factor "a" "Shall" designates the most important weighting level, the mandatory requirements. Unless specifically tailored out or modified by the contract, they constitute the firm contractual compliance requirements. Any deviations from the contractually imposed mandatory requirements requires the approval of the contracting officer (see 6.2.3).

b. Weighting factor "b" "Shall, where practicable", designates the second weighting level. Alternative designs, items, or practices may be used for specific applications when the use of the alternative is substantiated by documented technical trade studies. These trade studies shall be made available for review when requested or provided to the government in accordance with the contract provisions. Unless required by other contract provisions, deviations, from the "shall, where practicable" requirements do not require approval of the contracting officer.

c. Weighting factor "c" "Preferred" or "should" designates the third weighting level. Unless required by other contract provisions, deviations from these preferred requirements do not require approval of the contracting officer and do not require documented technical substantiation.

d. Weighting factor "d" "May" designates the lowest weighting level. In some cases these "may" requirements are stated as examples of acceptable designs, items, and practices. Unless required by other contract provisions, deviations from the "may" requirements do not require approval of the contracting officer and do not require documented technical substantiation.

3.2 General Design Requirements

3.2.1 Selection of Parts, Materials, and Processes

Unless otherwise specified in the contract, the parts, materials, and processes shall be selected and controlled in accordance with contractor established and documented procedures to satisfy the requirements specified herein. The selection and control procedures shall emphasize quality and reliability to meet the mission requirements and to minimize total life cycle costs for the applicable system. Electronic parts shall be JAN Class S or in accordance with ML-STD-1547. An additional objective in the selection of parts, materials, and processes shall be to maximize commonality and thereby minimize the variety of parts, related

tools, and test equipment required in the fabrication, installation, and maintenance of the vehicle. However, identical electrical connectors, identical fluid fittings, or other identical parts shall not be used on a space vehicle where inadvertent interchange of the items or interconnections could cause possible malfunction. The parts, and processes selected shall be of sufficient proven quality to allow the space equipment to meet the functional performance, reliability, life cycle, and strength requirements during its including all environmental degradation effects. Parts, materials, and processes shall be selected to ensure that any damage or deterioration from the space environment, or the outgassing effects in the space environment, will not reduce the performance of the space vehicle beyond the specified limits.

Materials Materials shall, where practicable, be selected that have demonstrated their suitability for the intended application. Inherently fungus-inert materials shall be used, where practicable. Combustible materials or materials that can generate toxic outgassing or toxic products of combustion shall not be used if cost-effective alternatives exist.

Materials shall, where practicable, be selected for low outgassing in accordance with SP-R-0022 (NASA JSC). The total mass loss in 24 hours shall be less than 1 percent, and the collected volatile condensable material shall be less than 0.1 percent when heated in vacuum (0.0001 Torr) to 125 deg C (257 deg F) and collected at 23 deg C (73 deg F). The hygroscopic nature of many materials such as ester lubricants, composites, electroformed nickel, and anodic coatings for aluminum should be recognized if they are used, since they emit water in a vacuum and therefore may be unsuitable for some applications. Analytical contamination models shall be used to evaluate performance impacts of outgassing on adjacent critical equipment. Swelling and shrinkage characteristics of materials shall be shown to be within acceptable tolerances under worst case humidity and temperature cycling.

Metals shall be corrosion resistant, or shall be suitably treated to resist corrosion when subjected to the specified environments. Protection of dissimilar metal combinations shall be in accordance with MIL-STD-889. Care shall be exercised in the selection of materials and processes in accordance with MSFC-SPEC-522 and MIL-STD-1568 to avoid stress corrosion cracking and brittle fracture failure modes, and to preclude failures induced by hydrogen embrittlement. For example, aluminum alloys 2020-T6, 7079-T6, and 7178-T6 shall not be used for structural applications. The use of 7075-T6, 2024-T3, 2024-T4, and 2014-T6 sheet material less than 6.3 millimeters (0.25 inches) thick is allowed, provided short transverse stresses (design, attachment, assembly, thermal, and residual) are below acceptable stress corrosion limits and corrosion protection systems are provided. Other forms of 7075 shall be heat treated to the -T73 or -T76 temper.

The following alloys and heat treatments shall not be used in applications where the temperature exceeds 65 deg C (149 deg F): 5083-H32; 5083-H38; 5086-H34; 5086-H38; 5456-H32; and 5456-H38. Heat treatment of aluminum alloy parts shall be in accordance with MIL-H-6088. Heat treatment of steel parts shall be in accordance with MIL-H-6875. Heat treatment of titanium and titanium alloys shall be in accordance with MIL-H-81200. All high strength steel parts heat treated at or above 1241 megapascals (180,000 pounds per square inch) ultimate tensile strength shall include appropriate test specimens from the same heat

of material as the part. These test specimens shall accompany the parts through the entire fabrication cycle to ensure that the desired properties are obtained. When acid cleaning baths or plating processes are used on high strength steel parts, the parts shall be baked at 190 deg C (374 deg F) for not less than 23 hours following such processes to alleviate hydrogen embrittlement problems.

Materials with low fracture toughness in the predicted operating environment, such as that exhibited by some plastics, shall not be used. Materials which are susceptible to cracking due to shock loads, or shock loads combined with low temperatures, shall not be utilized near any pyrotechnically actuated devices.

To minimize possible interactions with experiments or with the earth's magnetic field, magnetic materials shall be used only where necessary for equipment operation. For space vehicles that will operate at altitudes above 1000 kilometers (621 nautical miles), insulating materials or finishes having a resistivity greater than 10 megohm-meters shall not be used except in applications where charge buildup will not be excessive. Except for honeycomb panels, wall thickness of sheet metal and wall thickness on composite parts shall, where practicable, be greater than 0.5 millimeters (.020 inches) to minimize the possibility of handling damage. Honeycomb shall be vented to prevent excessive blow-off forces in vacuum.

Surface Treatments Any surface treatments or coatings used shall be such that completed components shall be resistant to corrosion. The design goal should be that there would be no harmful corrosion of the completed components or assemblies when exposed to moderately humid or mildly corrosive environments while unprotected during manufacture or handling, such as possible industrial environments or sea coast fog that could be expected prior to launch. Harmful corrosion shall be construed as being any type of corrosion which interferes with meeting the specified performance of the equipment or its associated parts. Protective methods and materials for cleaning, surface treatment, and applications of protective coatings shall be in accordance with MIL-F-7179. Cadmium and zinc coatings shall not be used. Chromium plating shall be in accordance with QQ-C-320. Nickel plating shall be in accordance with QQ-N-290 or MIL-C-26074. Anodic treatment of aluminum and aluminum alloys shall be in accordance with MIL-A-8625. Anodic treatment of magnesium shall be in accordance with MIL-M-45202. Corrosion protection of magnesium shall be in accordance with MIL-M-3171. Chemical films for aluminum and aluminum alloys shall be in accordance with MIL-C-5541.

3.2.2 Deployables

Deployables (see 6.2.6) shall, where practicable, be designed so that they are self supporting when placed in any orientation relative to gravity, while in either the stowed or deployed configuration. Deployables shall be designed with sufficient motive force to permit full operation during ground testing without depending upon the assistance of gravity to demonstrate deployment. Off-loading devices to simulate a zero-g condition are allowed so that the deployable does not have to act against the force of gravity. The deployables shall utilize redundancy, where practicable, in the design to improve the reliability of deployment. Techniques such as redundant springs or the use of a back up deployment device shall be considered to satisfy this requirement. Where a device rotates after deployment, as in sun

tracking solar arrays, the design shall be, where practicable, such that the center of gravity of the device is coincident with the axis of rotation to preclude the need for counterbalancing during functional testing. For all deployables, the deployment motion shall, where practicable, be controlled or restrained for the full range of travel. A controlled deployment may be provided by a motor and gear drive mechanism or other suitable means. A restrained deployment may be provided by a spring and damper drive mechanism or other suitable means. For all deployables, rebound of the deployable after contacting the stops should be minimized. Kick-off springs or other suitable devices shall, where practicable, be used to initiate deployment motion. These springs shall have a stroke long enough to ensure complete disengagement of the deployable from its retention mechanism. Field joints shall be provided, where practicable, to permit disassembly of all deployables from the space vehicle and to permit disassembly of rotating or translating elements from the remainder of the deployable to facilitate testing or replacement of parts. Deployables which require discrete sequential motions, each initiated by separate pyrotechnic events, shall have either the primary initiation of each event separately ground commanded or shall have a backup ground command for each event. Where cables (non-electrical) are used in the deployment mechanism, they shall be adequately guided and supported by fairleads. All pulleys shall utilize pulley guards which extend to the tangency points of the cable. Where operational acceleration loads are applied to a deployed assembly, such as those generated by space vehicle spin or propulsion forces, the design should preferably be such that the accelerations tend to drive the deployable into its deployed latched position.

Sequential Deployment Where two or more deployables are used on a space vehicle, the deployment of one shall not be dependent upon the successful deployment of the other, unless this sequential method of operation is unavoidable. Where this sequential dependency cannot be avoided, and the deployable that could be obstructed is mission critical (see 6.2.41, a fully redundant release device and fully redundant or backup deployment mechanism shall be provided for the deployable which, by its failure (see 6.2.8) could prevent successful deployment of the mission critical deployable.

Retention and Release Devices

General Positive retention provisions shall be provided for deployables in the stowed and in the deployed positions. The effects of deflections such as those induced by centrifugal forces or differential thermal growth of any deployable with respect to its space vehicle attachments shall be considered in the design of the attachments. Devices that may be subject to binding due to misalignment, adverse tolerances, or contamination shall not be used. Slip joints shall be avoided, where practicable. Flexures, four bar linkages, or other types of pivotal linkage are preferred. Self-aligning features, such as self-aligning bearings and rod ends, shall be used, where practicable, to preclude binding of pivoting elements. Continuous hinges, such as piano hinges, shall not be used for large deployable panels. Pyrotechnically actuated devices, motor driven devices, or other suitable techniques may be used to retain the deployable in the stowed position.

n . Release mechanisms which permit ejection of parts away from the space vehicle shall be avoided, where practicable. The design of latching devices shall be such that peaking of resistance near the end of travel of the deployable is minimized. Leaf spring latches which become the primary element for reacting deployment or deployment rebound loads shall be avoided, where practicable. Catches using a permanent magnet as the holding element shall be avoided, where practicable. The design and materials used for the retention devices shall be such that the stresses are maintained sufficiently below the fatigue endurance limit to avoid fatigue failures due to cyclic design load levels and environmental exposure. Retention and release devices shall be designed to preclude cold welding and friction welding (see 6.2.2). For surfaces that slide or separate during operational use, the contact pressures at the interfaces shall be minimized consistent with providing adequate ascent stiffness. These surfaces shall be fabricated from appropriate materials and lubricated so as to prevent galling or seizure. Where contact areas may be reduced from the nominal as a result of tolerance build-up, the minimum area which could occur shall be used in determining contact pressure.

Pin Pullers Where pin pullers are used, such as cartridge actuated or explosive pin pullers, they shall be designed to be in double shear. The design, installation, and check-out procedures for pin pullers shall ensure that loads due to misalignment of the pin are within design limits. A minimum retraction force margin of safety of 100 percent at worst case environmental conditions and under worst case tolerances shall be maintained for all nonexplosive pin pullers. Functional margins of cartridge actuated pin pullers shall comply with DOD-E-83578. Where the force margin of safety cannot be met with a release device, such as a pin puller, in the direct path of the primary load, designs which utilize the mechanical advantage of a linkage to reduce the loads on the device may be used. Release devices, such as pin pullers, shall have sufficient stroke so that complete release is attained under worst case tolerances and environmental conditions. As a minimum, pin pullers shall retract at least 2 millimeters (.08 inches) beyond the point of release calculated under worst case dimensional tolerances and environmental conditions. Redundant release devices shall be used, where practicable. An example is a toggle release mechanism where activation of either pin puller permits release.

Cartridge Actuated Devices Cartridge actuated device designs and their application shall comply with DOD-EL83578. Redundant pyrotechnic devices shall be used, where practicable.

Test Fixtures The fixtures to be used to assist in the fabrication and testing of the deployables shall be designed in conjunction with the deployable device to ensure compatibility. Where it is impractical to design the deployable to support itself in a one "g" field, supplemental supports may be employed for test purposes. These supplemental supports shall be designed so that their influence on the operation of the mechanism is minimized. The test fixtures shall incorporate, where practicable, provisions to measure torque vs angle and time vs angle, or equivalent linear measurements for linear devices. Where practicable, test fixtures shall have interlocks or other safety features incorporated to preclude damage to

the moving mechanical assembly or deployable in the event of testing malfunctions. Tethers and supports shall be incorporated as backup features to preclude damage in the event of failure.

3.2.3 Bearings

For deployables, hinges, and linkages, self-aligning bearings shall be used, where practicable, to preclude binding due to misalignments. Bearings shall not be used for ground current return paths or to carry electrical current. All ferrous material bearings shall employ, where practicable, a corrosion-resistant steel that is in accordance with QQ-S-763. Rolling element bearings shall, where practicable, be of 440C stainless steel; however, 52100 or MS0 steels may be employed providing they are suitably protected from corrosion (see 3.2.4). Rolling element bearings shall have a minimum hardness of Rockwell C58. Ball bearings used in applications where low torque ripple is required or where fatigue life is critical shall, where practicable, be fabricated of consumable electrode vacuum melted (CEVM) material.

Ball Bearings Bearings used in critical applications such as reaction wheels, control moment gyros, gimbals, de-spin mechanisms, and pointing devices shall meet ABEC 7, 7P or 7T tolerance or better in accordance with the AFBMA Standards. Nonstandard bearings or thin sectioned bearings where AFBMA tolerances do not apply shall have the manufacturers precision level most nearly equivalent to ABEC 7. Bearings used in noncritical applications such as one time use deployables, shall, where practicable, meet ABEC 5 or better. Stainless steel bearings shall, where practicable, also be in accordance with MIL-B-81793.

Design and Selection Ball bearing selection, mounting, and preloading shall be based on the following design considerations:

- a. Maximum combined axial, radial, and moment loads sustained during ground handling, launch, on orbit, or other operational modes

- b. Stiffness requirements

- c. Effects of temperature, temperature gradients, fits, tolerances, and initial preload on torque, stiffness, and life

- d. Lubrication

- e. Wear

- f. Smoothness of operation (torque ripple)

- g. Friction torques, considering breakaway and running, in the installed state

- h. Reliability and life

The design of each ball bearing installation shall be substantiated by analysis and either development tests or previous usage. The materials, stresses, stiffness, fatigue life, preload, and possible binding under normal, as well as the most severe combined loading conditions, and other expected environmental conditions shall be considered. Alignment, fits, tolerances, thermal and load induced distortions, and other conditions shall be considered in determining preload variations. An analysis of the effects of fits and temperatures shall be performed where the preload value is critical to meet either life, friction, or stiffness requirements. Bearing fatigue life calculations shall be based on a survival probability of 99.95 percent when subjected to maximum time varying loads. Appropriate correction factors shall be included to account for the selected bearing

materials and lubrication parameters. Bearing retainer design shall provide an acceptable contribution to bearing friction torque and torque variation. For noncritical applications, such as most deployables, if nonquite running is acceptable, the mean Hertzian contact stress shall not exceed 2758 megapascals (400,000 psi) when subjected to the yield load. During operation, the mean Hertzian stress shall not exceed 2310 megapascals (335,000 psi). The bearings shall be designed to withstand the ultimate load without fracture. The yield and ultimate factors of safety shall be as specified in 3.4.1.5.

Bearing Applications Requiring Quiet Operation or Low Torque Ripple For these applications, in addition to the requirements of Paragraph 3.2.3.1.1 bearing races and balls shall be designed so that stress levels are below the levels that would cause unacceptable permanent deformation during application of ascent loads. Where ball bearing deformation is required to carry a portion or all of the space vehicle ascent loads, and where smoothness of operation is required on orbit, the mean Hertzian stress levels of the bearing steel shall not exceed 2310 megapascals (335,000 pounds per square inch) when subjected to the yield load. The upper and lower extremes of the contact ellipses shall be contained by the raceways. The stress and shoulder height requirements of the races shall be analyzed for both nominal and off-nominal bearing tolerances. analysis and for quality control, For subsequent these bearings shall be serialized and measurements critical to their operation shall be recorded. Depending on the application and its requirements, consideration shall be given to recording preload, low speed torque, inside diameter, outside diameter, angle, radial play or control curvatures, raceway roundness, ball size and roundness, raceway surface finishes, and hardness.

Reuse of Bearings The design of bearings that are to be used for a number of flights shall be substantiated by analysis and either development tests or previous usage.

Bearing Alignment Run-out tolerances on shaft and housing diameters and shoulders shall be established in accordance with the needs of the application and the sensitivity of the bearing to misalignment. Bearings requiring stringent control of alignment include:

- a. Thin section, large diameter bearings
- b. Preloaded bearings
- c. Bearings operating continuously at speed
- d. Bearings requiring low torque ripple
- e. Bearings that oscillate

Overall misalignment of inner plus outer rings of these bearings shall, where practicable, not exceed 0.3 milliradians. Other bearings shall, where practicable, have overall misalignment not exceeding 1 milliradian.

Other Bearing Types Selection of other bearing types such as roller, needle, or journal, shall be based on the following design considerations:

- a. Maximum combined axial, radial, and moment loads sustained during ground handling, launch, on orbit, or other operational modes
- b. Temperature excursions and thermal gradients

- C . Ef f e c t s of t e m p e r a t u r e o n f i t s a n d t o l e r a n c e s
- d . S t i f f n e s s
- e . L u b r i c a t i o n
- f . S m o o t h n e s s o f o p e r a t i o n
- g. F r i c t i o n t o r q u e s
- h. R e l i a b i l i t y o v e r t h e d e s i g n l i f e

The design of each bearing installation shall be substantiated by analysis and either development tests or previous usage. The materials, stresses, stiffness, fatigue life, preload, and possible binding under normal, as well as the most severe combined loading conditions, and other expected environmental conditions shall be considered.

3.2.4 Lubricants

Lubrication shall be provided by greases, liquids, solid film lubricants, or a combination of either grease or liquid with solid film lubricants, soft metallic films, or sacrificial cages (for ball bearings) for all contacting surfaces having relative motion. The selection of lubricants for moving mechanical assemblies shall be based on the following considerations:

- a. Coefficient of friction
- b. Lubrication property changes in storage or in a vacuum environment
- C . D e p l e t i o n a n d w e a r o u t
- d. Operating temperature limits
- e. Creep properties
- f . v i s c o s i t y v s t e m p e r a t u r e p r o p e r t i e s
- g. Pressure coefficient of viscosity (ball bearings)
- h. Outgassing that could cause contamination such as on optical or thermal control surfaces (see 3.2.1.1)
- i . C o r r o s i o n p r o t e c t i o n
- j. P o s s i b i l i t y o f p o l y m e r i z a t i o n , p a r t i c u l a r l y d u e t o h i g h c o n t a c t p r e s s u r e s o r c o n t a m i n a n t s .
- k. Protection against cold welding and friction welding
- l. Cleanliness
- m. Run-in requirements such as rate of speed, load, and time duration
- n. Any requirements for demonstration of the suitability of the lubricant in a simulated space environment
- o. Any requirements of other environments such as humidity and salt spray
- p. Compatibility of the lubricant with other materials, particularly other lubricants if used

The lubricant chosen shall not cause detrimental effects on the moving mechanical assembly during or after operation at ambient conditions. The vent paths from all liquid lubricants shall be designed such that the outgassing products do not impinge directly on critical surfaces. The selection of the lubricant for use in the critical moving mechanical assembly shall be based upon development tests of the lubricant that demonstrate the ability of the lubricant system to provide adequate lubrication under all specified operating conditions over the design lifetime. The lubricant selected, and the processes used in

cleaning parts and applying the lubricant shall be delineated. Lubricants used with M50 or 52100 gear materials shall include corrosion inhibitors which prevent corrosion when the assembly is humidity tested in accordance with Paragraph 6.4.9 of MIL-STD-1540.

3.2.5 Brush and Slip Ring Assemblies

Electrical brushes and slip ring assemblies shall utilize interface geometry and materials that have demonstrated satisfactory performance in ground testing or space applications with similar requirements. Brushes that are nominally all carbon shall not be used. Electrical brushes shall be lubricated either by dry film lubricant that is a composite with the brush material or by a liquid lubricant that is impregnated into the brush material or retained in the area of the brush. Electrical brush and slip ring design shall be based on, but not limited to, the following considerations:

- a. Brush and slip ring interface geometry
- b. Contact forces
- c. Wear rate of brushes and rings
- d. Contact force control over the service life
- e. Zero "g" effects
- f. Materials used for the brushes and the rings
- g. Surface speeds
- h. Outgassing
- i. Sealing provisions
- j. Control or containment of wear debris
- k. Debris barriers or traps to prevent ring-to-ring or ring-to-case shorts
- l. Electrical current density
- m. Voltage level
- n. Brush-to-ring contact resistance
- o. Brush-to-ring electrical noise
- p. Temperature
- q. Contact area
- r. Lubrication
- s. Prevention of corrosion between the ring and brush
- t. The ability to conduct accelerated testing

Design verification testing of brushes and commutator or slip ring assemblies shall be conducted to demonstrate that the lubricant is not significantly deteriorated or driven from the areas requiring lubrication by adverse thermal, gravity forces, or other conditions; and that wear rates, wear debris, electrical noise level, friction torque, and lubricant loss are compatible with design requirements.

3.2.6 Electrical and Electronic

Electronic equipment shall be in accordance with DOD-E-8983. Wiring external to equipment enclosures shall be in accordance with DOD-W-83575. Where wiring harnesses cross a moving or rotating interface, the installation drawings shall define dimensions including loop sizes and distances to attachments, if applicable. Attachment clamps shall be provided

sufficiently close to any loops so that movement of the harness into the path of motion of the moving mechanical assembly cannot occur under any conditions. Wire bundles crossing a moving or rotating interface shall not contain strain-energy elements specifically added to assist deployment. All metal parts of the moving mechanical assembly shall be appropriately grounded to vehicle structure so that charge build-up in space and shock hazards on the ground can be minimized. The total current drawn from the power source during each mode of operation shall be determined to preclude inadvertent electrical overloads. These determinations or measurements shall be made at the minimum, nominal, and maximum voltage of the power source, and under the most severe environmental conditions. Overload protection, when required, shall be a part of the space vehicle electrical power control subsystem. A capability shall be provided to reset power disconnects by ground command.

3.2.7 Electric Motors

Alternating current motors shall conform to MIL-M-13787. Direct current motors shall conform to MIL-M-8609 or MIL-M-13786. For long term space application, brushless motors are preferred for their durability. Applications for conventional brush commutated motors shall consider lubrication, wear, input voltage range, and electromagnetic interference compatibility. Where applicable, the motor and its controller should be designed to provide adequate position and rate feedback for a local closed loop control system.

For applications where the motor performance is critical to the mission success, the design shall be based on a complete motor characterization. The motor characterization shall include rotor inertia, friction and damping parameters, back-EMF constant or torque constant, time constant, torque characteristics, speed versus torque curves, thermal dissipation, temperature effects, and, where applicable, analysis to demonstrate adequate margin against back driving. For applications where the motor is integrated into a higher assembly, the motor characterization shall be performed at the motor level prior to the integration.

Where motors are operated at high duty cycles, the design of the motor mount shall accommodate the heat transfer from the motor case to surrounding structure.

Stepper Motors For applications where limited speed, inherent holding capability, or discrete step motion is required, brushless permanent magnet stepper motors are preferred. In cases where the magnetic holding capability of the motor is used to hold the load in position, the design shall be based on the detent torque at each discrete rotor position for both the clockwise and counterclockwise direction. If the magnetic detent torque of the stepper motor is marginal to hold the load, trickle current may be applied to the motor winding when it is not energized. The trickle current should be disconnected when the motor is energized to reduce its effect on running torque. Detent torque margin may be assessed by considering the approximately sinusoidally varying electrical torque over a 4 step interval. Detent torque should change only if there is a change in shaft position. Under dynamic conditions, step integrity rather than torque margin should be the design criteria used to accommodate the range of conditions desired (load inertia, compliance, and resistive torque). Where stepper motors are used to drive a load, pulsing the stepper motor at a repetition rate that is at or near the natural frequency of the load shall be avoided. Variable rate stepping should be avoided due to the difficulties in analyzing and

demonstrating performance. Where other methods are practicable, stepper motor detent torque should not be used as holding torque to prevent backdriving of redundant drives.

Torque Motors For applications where smooth operation, precise position control, high speed, or high torque is required, permanent magnet, brushless, direct current torque motors are preferred for the efficiency. For brushless torque motors, electronic commutators such as the Hall effect sensor, resolver, and optical encoder have been successfully used for space applications. Where smooth operation is required, cogging torque should be minimized. When a redundant motor is mounted on a common shaft with the primary motor, the effect of one motor failure on the other motor should be minimized.

Direct Current Brush Motors Direct current brush motors may require special design and brush selection to avoid arcing or dust generation in both partial and high vacuum, as well as during ground testing. In particular, selection of brush motors should be avoided where operation at very low temperatures is required or where service is other than intermittent. Care should be taken to avoid use of motors with insufficient commutation segments which usually result in high brush wear. Guidelines and requirements for brush assemblies are provided in Paragraph 3.2.5.

Other Motor Types Other motor types are not to be excluded, but their applicability for space use may be limited. For example, reluctance motors have a high weight to torque or power ratio, series wound direct current motors have low efficiency, and shunt direct current motors have low starting torque.

3.2.8 Springs

Helical springs shall be in accordance with MIL-S-13572 and MIL-STD-29. Helical compression springs are preferred to helical tension springs. Helical compression springs shall, where practicable, be enclosed or otherwise captivated to prevent buckling, and to provide motive power even if broken. The attachments for retaining leaf springs shall be designed to reduce stress concentrations by such features as rounding of sharp corners or keeping mounting holes away from highly stressed areas. Spring design shall consider fatigue life and the effects of temperature on spring performance. To reduce the possibility of a reduction of potential energy due to creep, springs which are stored in a stressed condition shall be designed to maintain stress levels below the material proportional limit. Redundant springs, capable of working independently, shall be used in all applications, where practicable. When minimum friction is desirable, springs shall be dry film lubricated. The dry film lubricant shall be selected to avoid corrosion, and shall not contain carbon or powdered metal.

3.2.9 Dampers

Dampers may be utilized to absorb energy of impact of deployable devices at the extremes of travel or to limit the rate of travel during the entire travel period. Deformable material,

viscous dampers, or eddy current dampers are preferred to coulomb friction type dampers, which accomplish energy dissipation by the rubbing of two surfaces. Redundant dampers should be incorporated or a backup mechanism provided, to prevent a damper from becoming a single point failure mode.

Deformable Material Dampers

Viscous Dampers

Eddy Current Dampers

3.2.10 Gears

All gears used in moving mechanical assemblies shall be in accordance with the standards of the American Gear Manufacturers Association. The lubrication requirements of Paragraph 3.2.4 are applicable to gears. Hunting tooth (see 6.2.10) gear ratios shall be used, where the application is appropriate, to distribute wear. For better protection of the gear teeth, the through hardness or surface hardness or both may be increased, and the surface finish of the teeth improved through grinding, honing, lapping, and pre-run-in. The through hardness may be increased by material or heat treatment changes. The surface hardness may be increased by nitriding, carburizing, induction hardening, or anodizing. Undercutting of spur gear pinions should be avoided. Spur gear designs which have greater recess action than approach action are preferred. Spur gear contact ratios should be greater than 1.4 for power transmission gearing. Cantilever gear shaft mounting should be avoided in order to reduce non-uniform tooth face load distribution.

Aluminum gears are not recommended, except for light duty, limited life application where tooth wear and the coefficient of thermal expansion can be accommodated, and where compatibility with the selected lubricant can be established. An anodization process may be used to improve wear resistance for acceptable aluminum gear applications, provided that contact stresses will not cause the coating to crack. Gear tooth patterns should be checked on first assembly to establish that the pattern is well centered over the tooth flank, and that edge loading is not present.

Gear Design and Selection

Harmonic Drives A harmonic drive is a unique form of gear transmission that provides high gear ratio and torque capability in a compact configuration with minimum position error. Typically, a harmonic drive consists of a circular plate, a wave generator bearing, and flexspline. The harmonic drive shall be designed so it is not subjected to loads that cause ratcheting. Each element shall meet the applicable requirements stated herein. For example, the wave generator bearing shall meet the requirements of Paragraph 3.2.3. For long life applications, sufficient lubricant shall be provided to the bearings and to the gear teeth to ensure proper operation throughout the life cycle.

Precision Gears For precision gears, such as in fine pointing mechanisms, anti-backlash gearing shall, where practicable, be used. For critical applications, AGMA quality level 12 or better should be considered (see AGMA Handbook). Where gears are required to be matched sets, they shall be identified and marked as such.

3.2.11 Fasteners and Locking

Fastening systems shall be in accordance with MIL-STD-1515. Threaded parts shall be in accordance with MIL-S-7742 or MIL-S-8879. A minimum engagement of five full threads is required for threaded attachments; or for through bolts, the threaded ends shall protrude a minimum of two full threads beyond the end of the nut. Screw sizes smaller than 3.6 millimeters (No. 8) in diameter shall be avoided, where practicable. Where there are areas which may be sensitive to debris generated during assembly of threaded parts, blind holes should be considered. Tolerances shall be controlled to prevent threaded parts from bottoming in blind holes. The types, sizes, and quantities of fasteners used should be minimized. Consideration should be given to the frequency of access or use when selecting the type of fastener.

Locking Devices Positive locking devices shall be used on all fasteners. Preferred positive locking devices are bent tab washers, cotter pins, safety wire, self-locking threads, or self-locking provisions by means of plastic material contained in the nut, bolt, or screw. Self-locking nuts are preferred to bolts or screws that contain plastic material for use as a locking device. Where other locking devices are practicable, locking compounds shall not be used on fasteners to provide locking. Also, locking compounds shall not be used in areas where excess compound could migrate to surfaces which must remain free to move. Self-locking devices which depend upon an interference fit between metallic threads shall be avoided, where practicable, in applications where particulate contamination may cause damage or degradation to the equipment or vehicle. Where preload in fasteners is critical, strain gauges, crush washers, or equivalent techniques shall be used, where practicable, in lieu of torque wrench setting of the preload. Safety wiring and cotter pins shall be in accordance with MS-33540. Drawings shall clearly depict the safety wiring method and configuration used. Through bolts or screws with locknuts are preferred to threaded inserts. Threaded inserts shall be used in applications that require tapped holes in aluminum, magnesium, plastic, or other materials that are susceptible to galling or thread damage. When self-locking features are used, the screw length shall be sufficient to fully engage the locking device with a minimum of two turns under worst case tolerances. When self-locking features are used, an allowable range of run-in torque, or the maximum number of reuses that would still ensure an adequate lock, shall be specified. Spring type or star type lock washers shall not be used. Adjustable fittings or mounting plates which use oversized holes or slotted holes to provide adjustment shall not be dependent upon friction between the fitting or mounting plate and the mounting surface to provide locking. Diamond type serrations shall not be used.

Pivots

Snap Rings

3.2.12 Stops

3.2.13 Structure

Nonmetallic Structural Materials

Fiber-reinforced Composite Material Properties

Adhesives and Polymers

Allowables for Nonmetallic Materials

3.2.14 Pressurized Components

Pressurized components shall be in accordance with MIL-STD-1522.

3.2.15 Instrumentation

Sufficient diagnostic instrumentation shall be developed and provided, where practicable, as part of the moving mechanical assembly to determine the mode of failure should an on-orbit failure occur. These may be such devices as strain gauges, temperature sensors, pressure transducers, position indicators, potentiometers, switches, tachometers, accelerometers, or current monitors. When an electrical motor, other than a stepper motor, is used, the motor current shall be instrumented so that torque can be determined during development, acceptance, and qualification testing. Deployables shall, where practicable, have stowed and deployed position switches to indicate both release of the deployable from its restraining mechanism and end of travel or *latching position. Potentiometers may be used to provide data on intermediate positions. Where switches are used as indicating devices for deployables, the design of the switch mounting and the switch orientation shall be such that misadjustment of the switch shall not prevent full travel of the deployable to its deployment stop. In no case shall the direction of actuation of a switch be the same as the direction of motion of the mechanism, i.e., motion directly into the switch so that the switch can be damaged by misadjustment. Cam operated switches using ramps are preferred where the final position of the switch on the ramp is incapable of depressing the switch further than its normal operating range. Where switches are used, levers or other suitable devices shall be provided to decrease the sensitivity to adjustment of the switch and to ensure that sufficient over travel is provided after actuation of the switch. Switches shall be hermetically sealed. Sufficient on-orbit instrumentation shall be provided to measure critical temperatures and to detect off-nominal thermal conditions.

3.2.16 Test Points and Test Parameters

Test points shall be provided to accommodate a continuity of critical test parameter measurements from component acceptance tests through subsystem tests, vehicle acceptance tests, prelaunch checkout, and on-orbit test measurements. Test points shall be provided,

where practicable, as telemetry points on connector pins. The test parameters shall be chosen to provide assurance of satisfactory equipment performance and to isolate faults should they occur. The parameter test limits shall be established such that the measurements are made to an expanding accuracy tolerance that avoids the possible rejection of equipment which has passed tests conducted at lower levels of assembly. The on-orbit instrumentation and measurement techniques shall be used during ground tests to provide a data base that would permit parameter traceability with respect to variations in environmental conditions.

3.3 Performance

3.3.1 Error Budget for Precision Control Assemblies

For moving mechanical assemblies used in pointing applications where precision control is necessary, the error budget shall include performance errors due to misalignments, deflections, dynamic loads, thermal distortions, control system transients, steady state errors, friction, friction noise (friction or torque variations), friction hysteresis, structural and mechanical hysteresis, backlash, drive motor ripple, and other error sources as applicable.

3.3.2 Failure Modes and Effects

Required performance and reliability shall be ensured based upon a failure modes, effects, and criticality analysis (see 6.2.9). The failures to be considered shall include, but are not limited to, power outage, low voltage conditions, damper leakage, binding or excessive frictional loads, increase in friction noise, overtemperature conditions, excessive temperature gradients, failure of limit switches, partial or complete deployment failure, and structural failure in the moving mechanical assembly. During the preliminary design phase, an initial failure mode analysis shall consider providing redundancy in the design to achieve the required reliability. Failure modes established for the moving mechanical assemblies shall be used as a basis for selecting in-flight telemetry monitoring points to be used for ground diagnostic analysis should an anomalous condition occur.

3.3.3 Single Point Failures

Single point failure modes for moving mechanical assemblies* and their component parts shall be avoided, where practicable. This requirement may be implemented by such means as use of redundant springs or motors in deployment mechanisms so that failure of one spring or motor does not prevent full deployment and latching. Where redundancy is provided, the redundant portion of the moving mechanical assembly shall be in accordance with all the requirements of this specification. Where single point failure modes are unavoidable, or their avoidance is not practicable, the contractor shall ensure a satisfactory design based on an assessment of the mishap risk, and appropriate substantiating analyses and tests. The assessment and analyses shall include:

3.3.4 Tolerances

3.4 Environmental Design Requirements

3.4.1 Launch Environment

The equipment shall be designed to function within performance specifications after exposure in the launch configuration to the maximum launch or other nonorbital service environments. These environments include temperature, vibration, acoustic noise, shock, acceleration, atmospheric pressure, humidity, and electrical storms with associated radiation.

3.4.2 On-Orbit Environment

The equipment shall be designed to function within performance specifications following or, if appropriate, during exposure to environmental levels that exceed the maximum predicted on-orbit environments by the design factor of safety or design margin. These environments include thermal, vacuum, radiation exposure, shock, random vibration, and acceleration.

3.4.3 Fabrication, Storage, Transportation, and Handling Environments

3.5 Identification and Marking

Each moving mechanical assembly and interchangeable subassembly shall be identified by a nameplate. The nameplate identification may be attached to, etched in, or marked directly on the item where size and noninterference of operation permits. The nameplate shall utilize suitable letter size and contrasting colors, contrasting surface finishes, or other techniques to provide identification that is readily legible. The nameplate shall be capable of withstanding cleaning procedures and environmental exposures anticipated during the service life of the item without becoming illegible. Metal foil nameplates may be applied if they can be placed in an area where they cannot interfere with proper operation should they inadvertently become detached. Metal stamping shall not be used. Where practicable, identification nameplates on components and subassemblies shall be in locations which permit observation of the marking at the next higher level of assembly. Nameplates shall contain, as a minimum, the following:

- a. Item identification number
- b. Serial number
- c. Lot number
- d. Manufacturer
- e. Nomenclature

The marking of any two or more items intended for space applications with the same item number or identification shall indicate that they may be capable of being changed, one for another, without alteration of the items themselves or of adjoining equipment if the items also meet the specified flight accreditation requirements.

3.5.1 Data Cards

When size limitations, cost, or other considerations preclude marking all applicable information on an item, the nameplate may simply provide a reference key to cards or documents where the omitted nameplate information may be found. A copy of the referenced nameplate information or card shall accompany the item or assembly containing the item during ground tests and ground operations.

3.5.2 "NOT FOR FLIGHT" Markings

Items which by intent or by material disposition are not suitable for use in flight, and which could be accidentally substituted for flight or flight spare hardware, shall be red tagged or striped with red paint, or both, to prevent such substitution. The red tag shall be conspicuous and marked "NOT FOR FLIGHT." The red paint shall be material compatible and the stripes unmistakable.

3.6 Interface Requirements

3.6.1 Clearance

The clearance requirements between the moving mechanical assembly or deployable and any other structure or component shall be established and maintained. The manufacturing, assembly, and alignment tolerances as well as environmental conditions such as temperature, temperature gradients, vibration, distortion due to relaxation of the "g" field, effects of centrifugal forces, and acceleration during the critical periods of launch and on-orbit operations shall be taken into consideration for establishing clearance adequacy. The established clearances shall be maintained during transportation and all operational modes of the space vehicle. Obscuration of the operational fields of view of vehicle sensors by the moving mechanical assemblies shall be avoided except - where obscuration is a design requirement of the sensor system. Interface control drawings or layouts shall indicate the stowed, extended, and critical intermediate positions of the moving mechanical assemblies and deployables with respect to fields of view and surrounding structure or components.

3.6.2 Alignments

Provisions shall be included to permit of moving mechanical assemblies with respect to the space vehicle and other subsystem elements where appropriate. A shimming-type alignment technique may be used where additional surface area is necessary to provide a bearing surface or where electrical bonding is required. Where an adjustable-type alignment technique is used, locking provisions as specified herein shall be provided. Installation of alignment pins is permissible if the pins are positively retained with the moving mechanical assembly mounting face structure and permit the moving mechanical assembly to be easily removed from the vehicle.

3.6.3 Structural

In selecting the design for the structural interface of the moving mechanical assembly with the space vehicle, preference shall be given to simple approaches that minimize the complexity of the interface. The moving mechanical assembly shall, where practicable, be easily installed and readily accessible for inspection or removal. Standard wrench clearances shall be provided. The use of special tools shall be minimized. In addition, where complex connections and structural assembly interfaces are employed, the design of fittings shall be based on a minimum fitting factor of 1.15 at ultimate load.

3.6.4 Mechanical Interfaces

Where practicable, a common interface drill template shall be used to ensure correct mechanical mating, particularly for interfaces external to the moving mechanical assembly.

3.6.5 Interchangeability

Any two or more moving mechanical assemblies bearing the same part number or identification shall possess such functional and physical characteristics as to be equivalent in performance and durability and shall be capable of being changed, one for another, without alterations of the items themselves or of adjoining items except for alignment adjustments.

3.7 Operability

3.7.1 Reliability

The reliability allocations to the moving mechanical assemblies and the assigned design requirements shall ensure that the overall vehicle reliability requirements are met under the severe extremes of acceptance testing, storage, transportation, preflight testing, and operational environments. Highly reliable assemblies are usually obtained by elimination of single point failure modes unless it can be shown that the addition of redundancy actually reduces overall reliability due to added complexity. The incorporation of redundancy is usually an acceptable technique for meeting the reliability requirements. For designs that switch redundant subassemblies or units autonomously, or by ground command, the failure rates for the switching circuits, and for the redundant equipment while in the off line mode, shall be appropriately included in the reliability determination. The reliability of each moving mechanical assembly shall be analytically determined using piece part or component failure rates obtained from actual usage data where available, or evaluated at anticipated operating temperatures from standard data sources such as MIL-HDBK-217. The assembly reliability shall be evaluated in terms of events and usage cycles that occur in operation. The design margins shall preclude wear out of components within the service life of the space vehicle at the programmed duty cycle, including the wear associated with all preflight testing.

3.7.2 Maintainability

Except for possible relubrication after extended storage, or at prescribed intervals in multiple reuse applications, the moving mechanical assemblies shall, where practicable, be designed

so as not to require any scheduled maintenance or repair during their service life. Suitable development tests and inspections at appropriate intervals shall validate that scheduled maintenance is not required. However, access shall be provided for replacement of aged elastomeric materials which may have “expiration of useful life” dates that can occur before scheduled flight.

3.7.3 Human Engineering

Throughout the design and development of the moving mechanical assemblies, criteria shall be judiciously applied to obtain effective, compatible, and safe man-equipment interactions. The design of all moving mechanical assemblies shall be such that they may be tested or inspected to ensure that they are assembled and installed correctly. Provisions such as tabs, shoulders, and different thread sizes shall be employed to prevent assembly in any incorrect manner which may impair the intended functions of the moving mechanical assembly.

3.7.4 Service Life

The combined operational and nonoperational service life of the moving mechanical assemblies shall exceed the service life of the space vehicle in which they are mounted. The service life includes all operational and nonoperational time required or allowed after successful completion of acceptance tests.

3.7.5 Safety

General Safety

Space Transportation System Payloads

3.8 Manufacturing

3.8.1 Process and Controls

Acceptance and flight certification of space equipment is based primarily on an evaluation of data from the manufacturing process. Handling and storage of parts and materials for space applications shall be controlled consistent with the criticality of the end use of each item. Controlled environment and controlled access shall, be used to the extent required to avoid degradation of the quality and reliability of the parts, materials, and assemblies. The manufacturing process shall be accomplished in accordance with documented procedures and process controls which ensure the reliability and quality required for the mission. These manufacturing procedures and process controls shall be documented to give visibility to the procedures and specifications by which all processes, operations, inspections, and tests are to be accomplished by the contractor. This internal contractor documentation shall include the name of each part or component, each material required, the point it enters the manufacturing flow, and the controlling specification or drawing. The documentation shall indicate required tooling, facilities, and test equipment; the

manufacturing check points and pertinent process control thresholds; the quality assurance verification points; and the verification procedures corresponding to each applicable process or material listed. The specifications, procedures, drawings, and supporting documentation shall reflect the specifications in effect at the time the items were produced. These flow charts and the referenced specifications, procedures, drawings, and supporting documentation become the manufacturing process control baseline and shall be retained by the contractor for reference. It is recognized that many factors may warrant making changes to this documented baseline; however, all changes to the baseline processes used, or the baseline documents used, shall be recorded by the contractor following establishment of the manufacturing baseline or following the manufacture of the first item or lot of items. These changes provide the basis for flight accreditation of the items manufactured or of subsequent flight items.

The manufacturing process and control documents shall provide a contractor-controlled baseline that ensures that any subsequent failure or discrepancy analysis that may be required can identify the specific manufacturing materials and processes that were used for each item. In this way, changes can be incorporated into a known baseline item to correct problems.

3.8.2 Production Lots

To the extent practicable, parts for use in space equipment shall be grouped together in individual production lots during the various stages of their manufacture to ensure that all devices assembled during the same time period use the same materials, tools, methods, and controls. Parts and devices for space equipment that cannot be adequately tested after assembly without destruction of the item, such as explosive ordnance devices, shall have lot controls implemented during their manufacture to ensure a uniform quality and reliability level of the entire lot. Each lot shall be manufactured, tested, and stored as a single batch. Sequential lot numbers that indicate the date of manufacture shall be assigned to each production lot. (Typically, use three digits for the day of the year and two digits for the year.)

3.8.3 Contamination

Fabrication and Handling Fabrication and handling of space equipment shall be accomplished in a clean environment. Attention shall be given to avoiding nonparticulate (chemical) as well as particulate air contamination. To avoid safety and contamination problems, the use of liquids shall be minimized in areas where initiators, explosive bolts, or any loaded explosive devices are exposed.

Device Cleanliness The particulate cleanliness of internal moving subassemblies shall be maintained to at least level 500 as defined in MIL-STD-1246. External surfaces shall be visibly clean. The allowable product nonvolatile residue (NVR) level shall be maintained to at least level "G", as defined in MIL-STD-1246. Specific cleanliness requirements shall be determined for each moving mechanical assembly based on an analysis of overall system cleanliness requirements.

4 Quality Assurance Provisions

4.1 Responsibility for Inspections and Tests

Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection and test requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection and test requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for Compliance

All items shall meet all requirements of Sections 3 and 4. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract.

4.1.2 Test Equipment and Inspection Facilities

The manufacturer shall ensure that test and inspection facilities of sufficient accuracy, quality, and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of Inspections and Tests

The tests and inspections specified herein are classified as follows:

- a. Parts, materials, and process controls (see 4.3)
- b. Development tests (See 4.4)
- c. First assembly inspection (see 4.5)
- d. Component and subsystem level acceptance tests (see 4.6)
- e. Qualification tests (see 4.7)
- f. Vehicle level acceptance tests (see 4.8)
- g. Prelaunch validation tests and inspections (see 4.9)

4.3 Parts, Materials, and Process Controls

To ensure that a reliable moving mechanical assembly is fabricated, all parts and materials shall be adequately controlled and inspected prior to assembly. During fabrication, the tools and processes, as well as parts and materials, shall be adequately controlled and inspected to ensure compliance with the approved manufacturing processes and controls.

4.3.1 Records

Records documenting the accreditation status of the moving mechanical assemblies shall be maintained following assignment of serial numbers. Each moving mechanical assembly shall have inspection records and test records maintained by serial number to provide traceability from system usage to production lot data for the device. Complete records shall be maintained and shall be available for review during the service life of the system. The records shall indicate all relevant test data, all rework or modifications, and all installation and removals for whatever reason.

4.3.2 Manufacturing Screens

Each moving mechanical assembly and each critical subassembly shall be subjected to in-process manufacturing and assembly screens to ensure compliance with the specified requirements to the extent practicable. Compliance with the documented process controls, documented screening requirements, required hardware configuration, and general workmanship requirements shall be verified. At each level of assembly, each completed unit shall be subjected to visual inspection to ensure that it is free of obvious defects and is within Specified physical limits.

4.3.3 Nonconforming Material

Nonconforming material, components, or assemblies that do not meet the established tolerance limits set for the acceptance limits in the in-process screens shall be rejected for use. Any rejected material, component, or assembly may be reworked and rescreened in accordance with established procedures if system reliability is not jeopardized, and if the rework is not so extensive as to jeopardize the lot identity of the material or assembled unit.

If reworked material or an assembled unit that was lot controlled subsequently pass the in-process screens, it can again be considered part of the lot. Reassignment of units that were lot controlled to a different lot shall not be made. Nonconforming material or assembled units that do not satisfy these rework criteria shall be considered scrap.

4.3.4 Inspection of Parts

As a minimum, all castings, fusion welds, fibre composite or honeycomb parts and surfaces of stressed parts shall be inspected as follows:

a. Internal . All castings and all fusion welds shall be inspected for the presence of internal defects by radiographic methods in accordance with MIL-STD-453. Plastic composite structural elements shall be inspected for the presence of internal defects by an appropriate test method.

b. Surface defects. The surfaces of stressed parts shall be inspected in accordance with MIL-X-6868 (magnetic particle inspection method), in accordance with MIL-STD-6866 (liquid penetrant inspection method), or in accordance with other inspection methods, as appropriate. Heat-treated steel parts with tensile strength above 1,790 megapascals (259,617 pounds per square inch) ultimate that require a grinding operation after final heat treatment shall be inspected by nital etch on all surfaces subject to grinding.

All parts used in critical moving mechanical assemblies or parts comprising single point failures shall be completely inspected to ensure that no defects exist which could lead to degraded performance or failure.

4.3.5 Inspection of Assemblies

The dimensions, weight, finish, identification markings, and cleanliness of each moving mechanical assembly shall be inspected, as appropriate, prior to acceptance testing and prior to installation on the vehicle. Inspection of the assembly shall be made to ensure that a minimum thread engagement of five full threads exists for all threaded attachments; or where through bolts are used, that a minimum of two full threads protrude beyond the end of the nut. Fasteners shall be inspected for compliance with required torque levels. These inspections shall be made concurrently with the assembly of the parts. Pin pullers shall be inspected at the vehicle test level to verify adequate travel and travel margin. All moving mechanical assemblies shall be inspected before and after exposure to environmental tests at the component test level, at the vehicle test level, and at the launch site. In addition, the inspections should be conducted at intermediate test points. These inspection processes shall include, but are not limited to, the following:

a. Scratchefi or dammed surfaces . If surface damage is sufficient to affect the intended function of the assembly, or the performance is degraded (e.g., damaged thermal control surfaces, increased friction of contacting surfaces or contamination due to debris), the moving mechanical assembly shall be rejected.

b. Rust or corrosion Any parts exhibiting rust or cor ros ion shal l b; r e j e c t e d .

c. Fastener torque, Fasteners shall be visually inspected for looseness. In addi t ion, the torque shall be checked on preselected fasteners before and after exposure to each environmental test condition.

d. Handling damage. Graphite epoxy and other composite parts shall be visually inspected for damage. Where tubular composite members are used, the internal surfaces shall, where practicable, be visually inspected for damage, by means of a borescope or by other appropriate means.

e. Cleanliness. Product particulate cleanliness shall not exceed ieve 500 as defined in MIL-STD-1246. Product nonvolatile residue (NVR) shall be in compliance with the applicable requirements specified for each assembly.

f. Wirina harness-. Wiring harnesses shall be inspected to verify that they are properly secured and to verify that the proper configuration is maintained, particularly in the area of rotating parts or joints.

g. Electrical and fluid connectors Dust caps shall be installed on unused connectois during shipment or test. Although unnecessary disconnecting and reconnecting of connectors shall be avoided, inspection for connector out-of-round and broken or loose electrical pins shall be made, where practicable.

h. Multilayer Insulation. Multilayer insulation should be inspected at the vehicle test level and after vehicle shipment for adequate clearance with respect to adjacent moving mechanical assemblies to ensure that movement of the assemblies will not be impeded during operation. In addition, where switches are used for indicating purposes, the clearance between the multilayer insulation and the switch shall be verified to ensure that proper operation of the

switch can occur. .

i. Safety wire Fasteners utilizing safety wire shall be inspected to ensure that the wire is properly installed and that such wire cannot result in interference with other parts of the moving mechanical assembly.

j. Alignment appropriate. Alignment shall be inspected as Alignment mirrors shall be inspected to verify freedom from damage or displacement. Alignments of deployables shall, where practicable, be checked in their deployed positions.

k. Switch= Indicating switches shall be inspected to ensure that they operate satisfactorily and that adequate overtravel exists. Switch wires shall be inspected to verify that they are free from damage or sharp bends. Switch actuation shall, where practicable, be verified by means of electrical continuity.

l. Brinelling. Highly loaded surfaces such as preload fittings or overcenter latches shall be inspected for evidence of Brinelling damage to surfaces in contact.

m. Springs. Where accessibility permits, the following inspections and measurements shall be made on springs: inspection for broken coils or other damage and measurement of spring force or torque and spring rate on the fully assembled moving mechanical assembly. The spring load at the installed position and the spring rate shall be measured before assembly, recorded, and the springs properly serialized. The spring load and the spring rate shall also be measured and recorded after spring assembly or installation, as appropriate. The location of each serialized spring on the moving mechanical assembly shall be recorded. On all springs comprising single point failures, the heat treat shall be checked, where practicable, using an unstressed portion of each spring, such as on the ground flats on the ends of compression springs. Where there is a danger of causing damage as a result of the heat treat test, a lot sampling technique may be used if approved by the contracting officer. Where practicable, actual spring position and deflected shape during the entire range of travel shall be checked to avoid off-nominal loading due to coil displacement and tang movement

n. Dampers Viscous dampers shall be performance and leak checked. Deformable material dampers shall be inspected to ensure that the material is not deformed or damaged.

o. Clearances shall be inspected to ensure compliance with the specified requirements.

p. Electric = Where accessibility permits, the following inspections and measurements shall be made on electric motors:

1. Verification of the part number.

2. Identification of special precautions (if any) such as use of keepers during armature removal, and observation of lead color code for polarity where suppression diodes are used.

3. Inspection for adequate lead length and clearance from other leads to which motor noise might introduce electromagnetic interference.

4. Adequate mount heat sinking.

5. Inspection of freedom of motion without binding or misalignment problems.

Operationally the input current may be monitored to verify brush and motor performance.

4.3.6 Lubricant Processing Procedures

Lubricant processing procedures shall be established to control the uniformity, repeatability, cleanliness, accountability, a

and amount of lubricant in the moving mechanical assemblies. These procedures shall also delineate the processes of removal of bearing shipping lubricants. Shipping lubricants shall be analyzed for compatibility with the applied lubricant. Porous ball bearing retainers and lubricant reservoirs which have shipping lubricants in them shall be replaced before processing with the flight lubricant. Each bearing to be processed with a liquid lubricant shall be tested for wettability prior to application of the flight lubricant. The wettability test method shall be approved by the contracting officer.

4.4 Development Tests

Development testing may be performed on moving mechanical assemblies to supplement analytical techniques in the qualification of design parameters and to promote the evolution of new design concepts. Developmental testing shall be used to validate new design concepts for critical components and assemblies as a prelude to qualification testing. The nature and extent of developmental testing on components, subsystems, and complete moving mechanical assemblies shall be sufficient to ensure that qualification testing of new designs will produce minimal failures which would then require redesign. Developmental testing shall be conducted using MIL-STD-1540 as a guide. Critical components requiring development tests should be identified early in the design and development program, and a list of tests and items so identified should be documented. The adequacy of the development testing is a contractor responsibility; however, some contracts may allow the contracting officer to delay approval to proceed with qualification testing pending a review of the development test results. In that case, a particular test should not be considered completed until a minimum of 30 days after the test results have been presented to the contracting officer. Unless the design is substantiated by previous usage, developmental testing shall include the following, as applicable:

- a. Tests to substantiate the design of each bearing installation with respect to materials, stresses, stiffness, fatigue life, preload, and possible binding under normal, as well as the most severe combined loading conditions, and the expected environmental conditions.

- b. For bearings that are reused, tests to characterize the wear and lubrication changes resulting from bearing reuse, and the effects these changes have on torque margin.

- c. Where practicable, tests to substantiate the effects of temperature gradients on bearing preload and stiffness.

- d. For critical moving mechanical assemblies, tests of the lubricant that demonstrate the ability of the lubricant system to provide adequate lubrication under all specified operating conditions over the design lifetime.

- e. Where contact pressure is sufficiently high, such that friction welding might be possible, design and process verification tests shall be conducted at the piece part or subassembly level to verify the selection of materials and lubricants in the proposed application.

- f. For brushes and commutator or slip ring assemblies, tests to demonstrate that the lubricant is not significantly deteriorated or driven from the areas requiring lubrication by adverse thermal, gravity forces, or other conditions; and that wear rates, wear debris, electrical noise level, friction torque, and lubricant loss are compatible with design requirements.

g. For deployable devices, minimum available driving capability and maximum load termination shall be verified through an appropriate test program; each element of resistance shall be characterized in this test program.

h. For moving mechanical assemblies driven by electrical motors, a torque versus current relationship for each motor under minimum, maximum, and ambient thermal and vacuum conditions shall be established.

i. Where practicable, tests to substantiate that scheduled maintenance is not required.

j. Eddy current dampers shall be tested for repeatability of damping rate and torque values for twice its expected operational life cycle. The eddy current damper shall be tested at temperature extremes to determine its damping rate over the temperature range.

4.5 First Assembly Inspection

An inspection of the first of each of the moving mechanical assemblies produced with production equipment and procedures shall be conducted at a time and location acceptable to the contracting officer. There shall be no discrepancies among the equipment, the fabrication tooling used, the released drawings, the test data, the inspection records, and the specification requirements. First assembly approval is valid only on the contract or purchase order under which it is granted, unless extended by the contracting officer to other contracts or purchase orders.

4.6 Component and Subsystem Level Acceptance Tests

The configuration and workmanship of the completed hardware shall be verified by inspection prior to the start of acceptance testing. Acceptance tests shall be conducted on each moving mechanical assembly in accordance with the acceptance test requirements of MIL-STD-1540. The acceptance tests shall incorporate the run-in, functional, and environmental test requirements stated herein. The acceptance tests shall be structured to detect workmanship defects that could affect operational performance. Components fabricated from fiber composite material or of honeycomb construction shall be subjected to acceptance proof testing. The acceptance testing of moving mechanical assemblies that are part of a deployable device shall, where practicable, be conducted with the moving mechanical assembly attached to the deployable device. In some cases, such as solar array drives, despin bearing assemblies, or pointing devices, dummy loads may be substituted for the driven member. The dummy loads shall provide a reasonable representation of the dynamic characteristics (such as inertia, stiffness, free play, and natural frequencies) of the actual driven member. The effect of friction due to the additional weight of the dummy load on the drive unit shall be accounted for in evaluating the test results. Dummy loads are also permitted to minimize the effects of air damping where large panels are deployed under ambient conditions. Appropriate quantitative measurements such as torque vs angle measurements and time vs angle measurements, or equivalent linear measurements for linear devices, shall be made during run-in, functional, and environmental acceptance testing of all moving mechanical assemblies. For spring driven devices, torque vs angle measurements shall be made during restowing as well as during deployment in order to generate a torque vs angle hysteresis curve to determine torque margin. For linear devices, equivalent linear measurements shall

be made to determine force margin. As a minimum, these measurements shall be made under ambient and acceptance level hot and cold temperatures. When an assembly is too large or complex to be tested, results of component or subassembly torque or force margin testing may be added to obtain the assembly torque margin. Moving mechanical assemblies that contain redundancy in their design shall, where practicable, demonstrate performance to their requirements in each redundant mode of operation. Functional operation and alignment of each moving mechanical assembly, including instrumentation that is an integral part of the assembly, shall be tested and checked, where practicable, before and after exposure to each environmental acceptance test. All swaged ends on rods or cables shall be 100 percent tested with a load equal to twice the limit load (see 6.2.12). (Note: the proof factor of safety shall be 2.0 for swaged ends on rods or cables.)

4.7 Run-in Test

After initial functional testing, a run-in test shall be performed on each moving mechanical assembly before it is subjected to further acceptance testing, unless it can be shown that this procedure would be detrimental to performance and would result in reduced reliability. The primary purpose of the run-in test is to detect material and workmanship defects which occur early in the component life. Another purpose of the run-in test is to wear-in parts of the moving mechanical assembly so that they perform in a consistent and controlled manner. Satisfactory wear-in may be manifested by a reduction in running friction to a constant low level. The run-in test shall be conducted for a minimum of 50 hours except for items where the number of cycles of operation, rather than hours of operation, is a more appropriate measure of the capability to perform in a consistent and controlled manner. For these units, the run-in test shall be for at least 15 cycles or 5 percent of the total expected service life cycles, whichever is greater. The run-in test conditions should be representative of the operational loads, speed, and environment; however, operation of the assembly at ambient conditions may be conducted if the test objectives can be met and the ambient environment will not degrade reliability or cause unacceptable changes to occur within the equipment such as the generation of excessive debris. During the run-in test, sufficient periodic measurements shall be made to indicate what conditions may be changing with time and what wear rate characteristics exist. Test procedures, test time, and criteria for performance adequacy shall be in accordance with a test plan that has been approved by the contracting officer. All gear trains using solid film lubricants shall, where practicable, be inspected and cleaned following the run-in test.

4.8 Functional and Environmental Acceptance

Each moving mechanical assembly shall be subjected to function;1 and environmental tests that are in accordance with the applicable acceptance test requirements of MfL-STD-1540. Functional tests shall be structured to demonstrate that the moving mechanical assembly is capable of operating in such a manner that all performance requirements are satisfied. Functional tests are required before and after exposure to environmental test conditions in order to establish whether damage or degradation in performance has occurred. These tests shall be sufficiently comprehensive and shall include sufficient measurements to determine

whether performance specifications are met. The functional tests are usually conducted at room ambient conditions, with the initial functional testing used as a baseline against which subsequent performance is compared. Where a device is designed to operate in extreme heat or cold or in some other environmental extreme, and a functional test at ambient conditions would not be significant, the functional tests shall be conducted in the appropriate environment that would demonstrate performance. All command functions should be exercised during functional testing. The moving mechanical assemblies shall be tested in their launch or in their on-orbit configurations corresponding to the environment being simulated. They shall be passive or operating corresponding to their state during the launch or on-orbit operational phases. Fully assembled deployables shall, where practicable, have torque vs angle and time vs angle measurements, or equivalent linear measurements for linear devices, made during thermal vacuum testing. When a harmonic drive is used for precision pointing applications, full functional testing should be performed to verify the harmonic drive characteristics at each level of assembly. Functional tests should include torsional stiffness, damping, and friction characteristics. After a harmonic drive is assembled into the parent assembly, it shall be inspected for the detoidal condition (see 6.2.5). When operation of a deployable in a vacuum chamber or measurements in a vacuum chamber are impractical, the fully assembled deployable shall have deployment tests, including torque vs angle and time vs angle measurements, or equivalent linear measurements for linear devices, made in a hot or cold box. When the fully assembled deployable is so large that it cannot be completely enclosed within a hot or cold box, the moving mechanical assemblies themselves may be separately enclosed within several smaller hot or cold boxes which surround only the moving mechanical assembly portion of the deployable device. Test procedures, test time, and criteria for performance adequacy shall be as approved by, the contracting officer. No disassembly, adjustments, or repair shall be made on moving mechanical assemblies during functional and environmental acceptance testing or after completion of testing unless approved by the contracting officer.

4.9 Qualification Tests

The qualification tests of the moving mechanical assemblies shall be conducted in accordance with the qualification test requirements of MXL-STD-1540. The 6 dB, 10 deg C (18 deg F), or other design factors of safety or margins specified herein include test condition tolerances that are those allowed in MIL-STD-1540. When the actual qualification or acceptance test tolerances can be shown to be less than those specified in MfL-STD-1540, the qualification test levels may be appropriately reduced in accordance with provisions specified in MXL-STD-1540. The qualification tests shall incorporate the design life verification test, functional test, and environmental test requirements stated herein. Moving mechanical assemblies which are required to operate more than once during their on-orbit life shall be tested using dedicated qualification items. Where a moving mechanical assembly is required to operate only once during its on-orbit life, consideration may be given to flight use of the qualification article based on the requirements of Section 8 of MIL-STD-1540. However, any flight use of a qualification article shall have prior approval by the contracting officer. The qualification tests shall be structured to demonstrate the design adequacy and design margins. An acceptance

test shall precede the qualification tests. Torque vs angle measurements and time vs angle measurements, or equivalent linear measurements for linear devices, shall be made during qualification testing of all moving mechanical assemblies. Satisfactory completion of these tests or equivalent proven space vehicle operational performance is required for flight certification or qualification. Compliance with all or portions of the qualification requirements based upon tests of similar items or upon equivalent space vehicle operational performance shall require approval of the contracting officer. The general test measurements and test configurations used for qualification tests shall be similar to those used for acceptance tests. Moving mechanical assemblies that contain redundancy in their design shall, where practicable, demonstrate performance to their requirements in each redundant mode of operation during the qualification test. Functional operation and alignment of each moving mechanical assembly, including instrumentation that is an integral part of the assembly, shall, where practicable, be tested and checked, before and after exposure to each qualification test.

4.10 Design Life Verification Tests

The design life verification tests are intended to evaluate lubricant suitability, release and deployment life cycle margins, wear life, and avoidance of fatigue. One or more items of each of the moving mechanical assemblies, produced with production equipment and procedures, shall be subjected to a design life verification test. The number of units to be tested should be sufficient to achieve the desired confidence level for the test results. These tests shall be conducted to simulate operational use within the range of the worst predicted operational environments. The items shall be subjected to tests which demonstrate the capability to perform the full operational life cycle. Masses, loads, and stiffness of support structure shall be actual or simulated operational values, and any interfacing equipment such as thermal heaters, cabling, or hoses shall be in place so that operational conditions are simulated. The test and criteria for performance adequacy shall be approved by the contracting officer. For items having a relatively low percentage duty cycle, it shall be acceptable to compress the operational cycle into a tolerable total test duration. For assemblies which operate continuously on-orbit, or at very high percentage duty cycles, accelerated test techniques may be employed if such an approach can be shown to be valid. For such assemblies, tests shall be under proper environmental conditions, of varying duration, followed by disassembly and inspection. A single item may be used provided that periodic disassembly and inspection does not influence the test conditions sufficiently to invalidate the test results. Detailed analyses of lubricant consumption, debris accumulation, wear, or other critical parameters may be used to provide bases for forecasting the expected life of the assembly. The moving mechanical assemblies used for life test shall be essentially identical with the flight items. The only differences between the test items and the flight items shall be those changes necessary for incorporation of test instrumentation. These differences shall not jeopardize the validity of the tests with respect to the flight hardware. Sufficient instrumentation shall be used to provide knowledge of the operating conditions within the assemblies during life testing such as internal pressures, temperatures, and temperature gradients. Where liquid lubrication is used, consideration shall be given to measuring lubricant loss, degradation, distribution, quantity, and outgassing constituents over the duration of the test. The instrumentation shall be similar to that employed in other test phases to provide a basis for comparison

of the design life test conditions with those of other selected ground tests and with those during orbital operations. The design life test moving mechanical assembly shall be operated as expected in flight with equipment operating in accordance with the predicted duty cycle. The tests shall include variations of the expected flight usage, such as power down modes to check low temperature operation of the system, to the degree practicable without exceeding the thermal limits of the equipment. All electrical slip ring and commutator assembly tests shall be conducted with representative levels of electrical current at the rated voltage across the interface. All moving mechanical assemblies shall be tested for at least twice the number of duty cycles expected in operational use, plus twice the number of duty cycles expected during component and vehicle functional and environmental tests. Stops shall be tested by intentionally running the moving mechanical assembly into the stops whether or not the moving mechanical assembly has limit switches to prevent contacting the stops in normal operation. The stops shall be tested for at least twice the number of duty cycles expected in operational use, plus twice the number of duty cycles expected during component and vehicle functional and environmental tests. For moving mechanical assemblies which employ limit switches and do not normally contact the stops, the qualification tests of the stops may be conducted as a separate subassembly level test with the switch inactive. A functional test shall be conducted after the design life verification test has been completed and the assembly shall be disassembled and inspected for anomalous conditions. The critical areas of parts which may be subject to fatigue failure shall be inspected to determine if failure has occurred.

4.11 Functional and Environmental Qualification

One item of each of the moving mechanical assemblies, produced with production equipment and procedures, shall be subjected to environmental qualification testing to verify satisfactory performance at the design environmental levels. Tests shall be in accordance with the applicable qualification test requirements of MIL-STD-1540. The moving mechanical assemblies shall be tested in their launch or in their on-orbit configurations corresponding to the environment being simulated. They shall be passive or operating corresponding to their state during the launch or on-orbit operational phases. Test procedures, test time, and criteria for performance adequacy shall be as approved by the contracting officer. Qualification testing of deployables shall include tests to simulate the lowest motive force combined with the highest resistance under the most adverse environmental conditions to provide the worst case torque margin. In addition, the highest motive force combined with the lowest resistance under the most aiding environmental conditions shall be tested to provide the worst case loading including the loads against the stops. Sufficient measurements shall be made to show that the requirements of this specification are met under both of the above conditions. Measurements shall include torque vs angle and time vs angle, or equivalent linear measurements for linear devices. For deployables, time correlated video recording, or motion picture film coverage, and potentiometers, accelerometers, or other appropriate instrumentation shall be utilized for all functional and environmental qualification tests to enable the dynamics of the deployment to be determined and to establish time vs angle or time vs distance acceptance test criteria limits. A one-time release test of pyrotechnic devices shall be conducted in the fully assembled configuration using a worst case lowest

explosive charge loading of 80%. This release test shall be conducted using worst case tolerances and worst case environmental conditions. To evaluate the possibility of structural damage to moving mechanical assemblies actuated by pyrotechnic devices, a one-time test shall also be conducted under worst case highest explosive charge loading. A 120 percent explosive charge load shall, where practicable, be used for this test. Instrumentation shall be provided on adjacent equipment during release tests to assess the transmitted shock loads. A static load test to demonstrate the required strength, safety factors, and margins shall be conducted. Following the static load test, the assemblies shall be completely disassembled and reinspected for possible damage.

4.12 Vehicle Level Acceptance Tests

The vehicle level acceptance tests shall be conducted in accordance with the requirements of MIL-STD-1540. After assembly of the deployable on the space vehicle, a time correlated video recording or motion picture film of deployment shall be utilized to ensure that the dynamics of deployment are within specified time vs angle or time vs distance pass-fail criteria. Where deployment is impractical after assembly on the space vehicle, this recording of the dynamics of deployment shall be made during the last functional deployment prior to mounting of the deployable on the space vehicle. A first motion cold test of all deployables shall be included, where practicable, as part of the space vehicle thermal testing to verify release of the deployables at the acceptance level cold temperature.

4.13 Prelaunch Validation Testing and Inspection

The prelaunch validation tests shall be conducted in accordance with the requirements of MIL-STD-1540.

4.14 Interface Tests

Where practicable, a complete deployment of deployable flight hardware in the flight configuration shall be conducted after attachment of the deployable to the space vehicle. A hand held walkout, if appropriate, is sufficient to satisfy this requirement. This test should be conducted at the latest possible time prior to flight. Where a complete deployment is not practical due to the size of the deployable, lack of a “one-g” supporting fixture, or risk of damage associated with a hand held walkout, an alignment check and an initial motion release test shall be conducted in lieu of the above requirements. A measurement of the minimum breakaway force shall be made to ensure that no binding of the tripping mechanism or excessive friction due to distortions of pivot support brackets is present.

4.15 Inspection

Where practicable, the inspections listed in Paragraph 4.3.5 shall be repeated at the launch site after assembly of the vehicle in the flight configuration. Deployments such as hand-held walkouts, or other deployments such as those required to provide access to nearby equipment, or as part of pre-flight checks, should not be performed without the presence of an inspector

to witness, approve, and sign-off acceptability of the handling, deployment, and restowing procedures.

4.16 Prelaunch Validation Exercises

Prelaunch validation exercises shall include contingency cases involving anomalous operation of moving mechanical assemblies. For example, partial deployment of space vehicle devices may cause additional effects on the space vehicle such as changes to the thermal control of the space vehicle, changes to the inertial properties of the space vehicle, degradation of communication systems, and possible blocking of sensors. In some cases the effects of the failure may be circumvented by operation of the space vehicle in a different mode, such as increasing the space vehicle spin speed to aid in deployment of a deployable device which has experienced a hang-up. These contingency plans should be formulated and exercised to the extent practicable.

4.17 Prelaunch Validation of Bearings that are Reused

Bearings that are to be used for a number of flights shall be inspected and retested after the initial flight. Individual bearing tests shall be conducted to determine if the resistive torques are within acceptable limits. If the mechanism torque or force margins are less than the minimum acceptance requirements, the bearings shall be completely inspected, refurbished, and retested as necessary after each flight. If the mechanism torque or force margins are considerably higher than minimum acceptance requirements, the bearings tests and inspections following subsequent flights may be relaxed.

4.18 Modification, Rework, and Retest

Completed moving mechanical assemblies shall be modified and reworked with the same high quality assurance provisions and criteria as the original assembly. Inspection and retesting shall be in accordance with MIL-STD-1540 and the requirements stated herein, or as directed by the contracting officer. Moving mechanical assemblies that have successfully completed the run-in test and are subsequently repaired, reworked, modified, or reacceptance tested shall not be given another run-in test unless the rework is so extensive as to invalidate the initial run-in. Moving mechanical assemblies that are not assembled on a space vehicle but are placed in storage for more than one year shall be retested in accordance with the established thermal or thermal vacuum acceptance test procedures prior to use. After two years of storage and every year thereafter, the liquid lubricants used in these moving mechanical assemblies shall, where practicable, be sample tested to evaluate degradation. If there is evidence of lubricant degradation, the cause of the degradation shall be eliminated and the assembly relubricated.

References

1. MA2-00-057 Mechanical System Safety, 2000.
2. NSTS/ISS 18798 Rev B Interpretations of NSTS/ISS Payload Safety Requirements, 1997.

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Appendix A: Original DFMR Requirements Text

1 Binding / Jamming / Siezing

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